6.3 Live Camera Feed

In this section, you can learn how to connect to the hotspot generated by the ESP32-Cam, and then log into a provided URL to view the live camera feed.

Note: if you are unable to use the live camera feed function or have downloaded other program firmwares, please reflash the image transmission firmware. This allows you to use the live camera feed function.

To download the firmware, please refer to "4. Image Postback Firmware Flashing (Optional)" in this section.

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1. ESP32-Cam Camera Connection

1.1 Connection Modes Introduction

ESP32-CAM has two kinds of network modes as follows:

① AP direct connection mode: the development board can open the hotspot to connect with the phone, but it cannot connect to an external network.

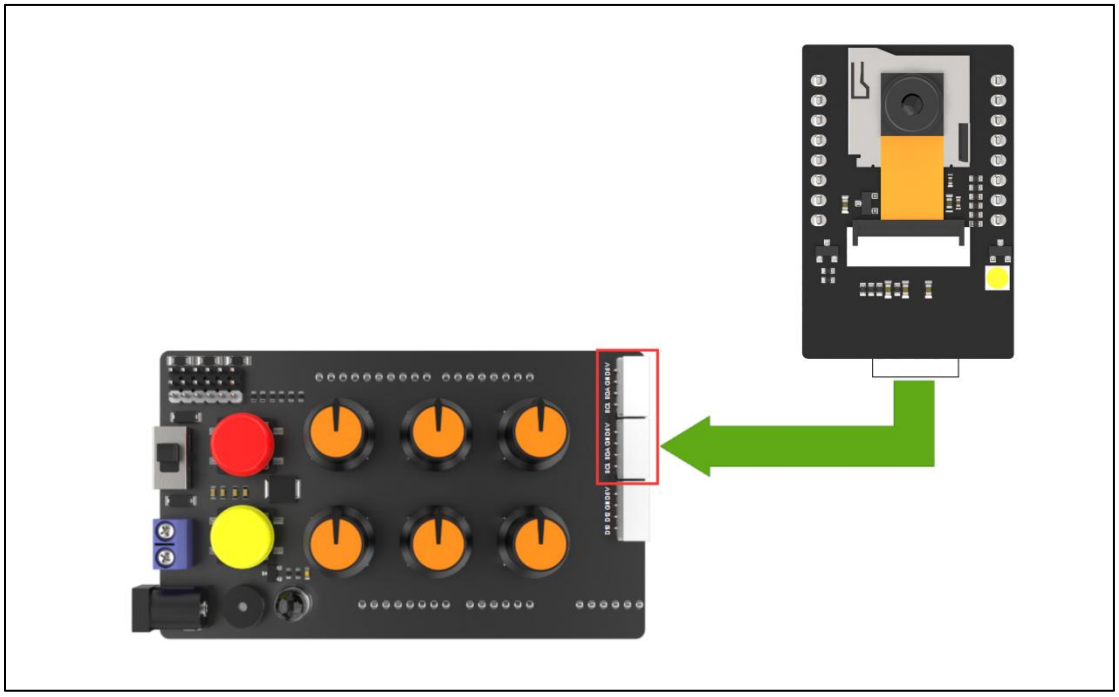
② STA LAN mode: the development board can actively connect to a designated hotspot or Wi-Fi, enabling communication with an external network.

The AP direct connection mode is more convenient to operate. We recommend you to first learn the configuration method of the direct connection mode to experience the corresponding functions. The STA LAN mode can be selected according to your needs.

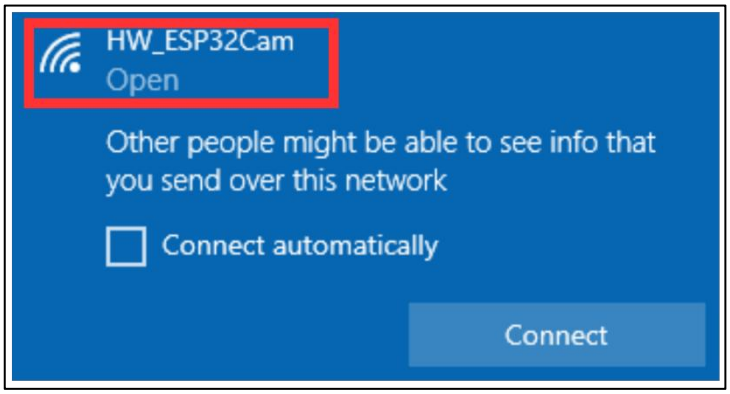
No matter which kind of mode you choose, the game function is the same.

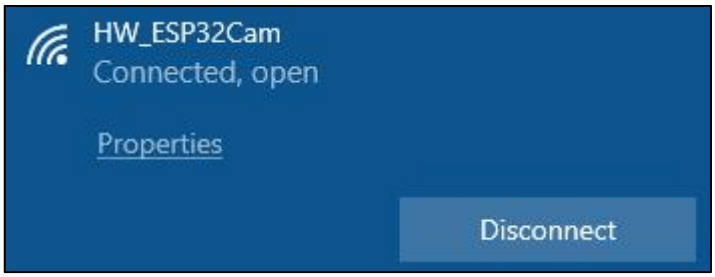
1.2 Direct Mode Connection (Must Read)

1) Make sure the camera is connected to the I2C interface on the expansion board and turn the robotic arm on.

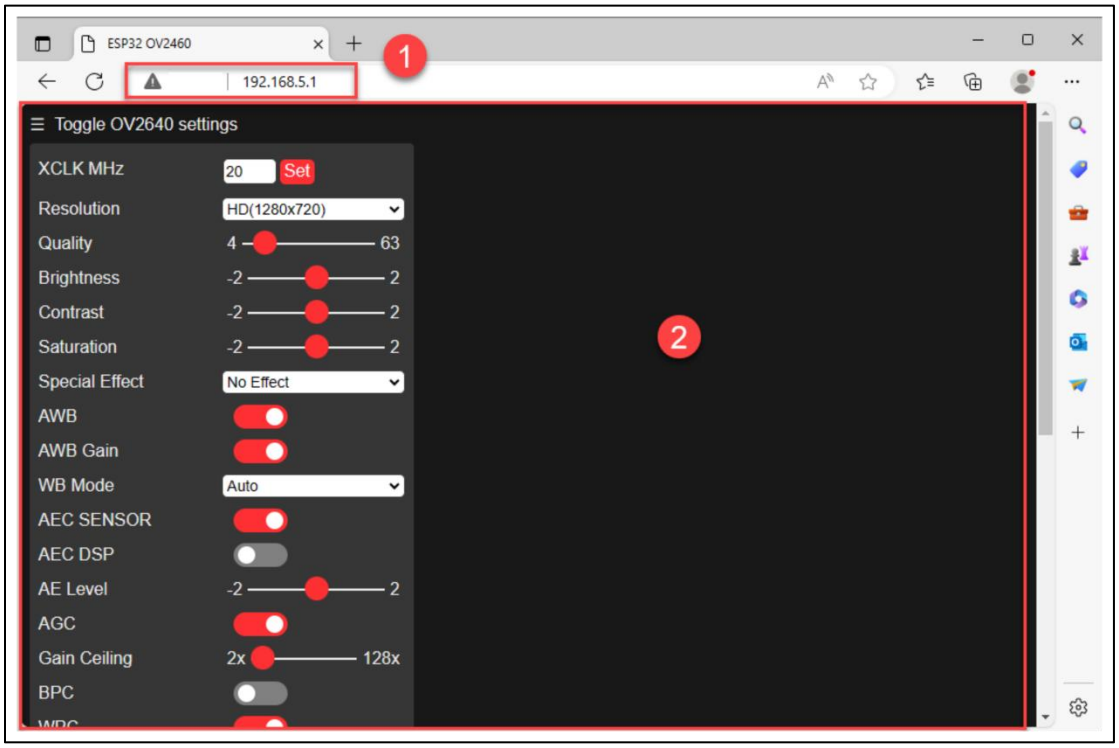


2) The camera will generate a hotspot named “HW_ESP32Cam”. Click to connect.



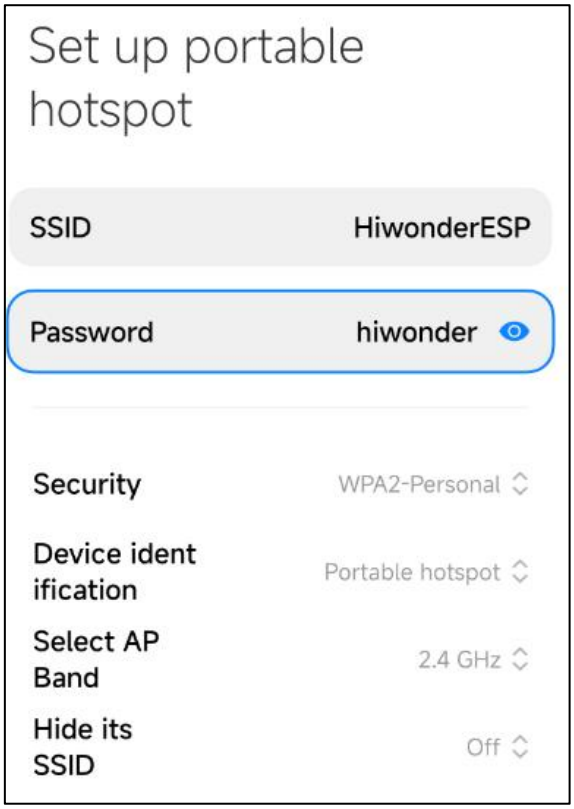


3) After the connection is completed, enter the address “192.168.5.1” in the browser and click the research to enter the camera transmission interface.

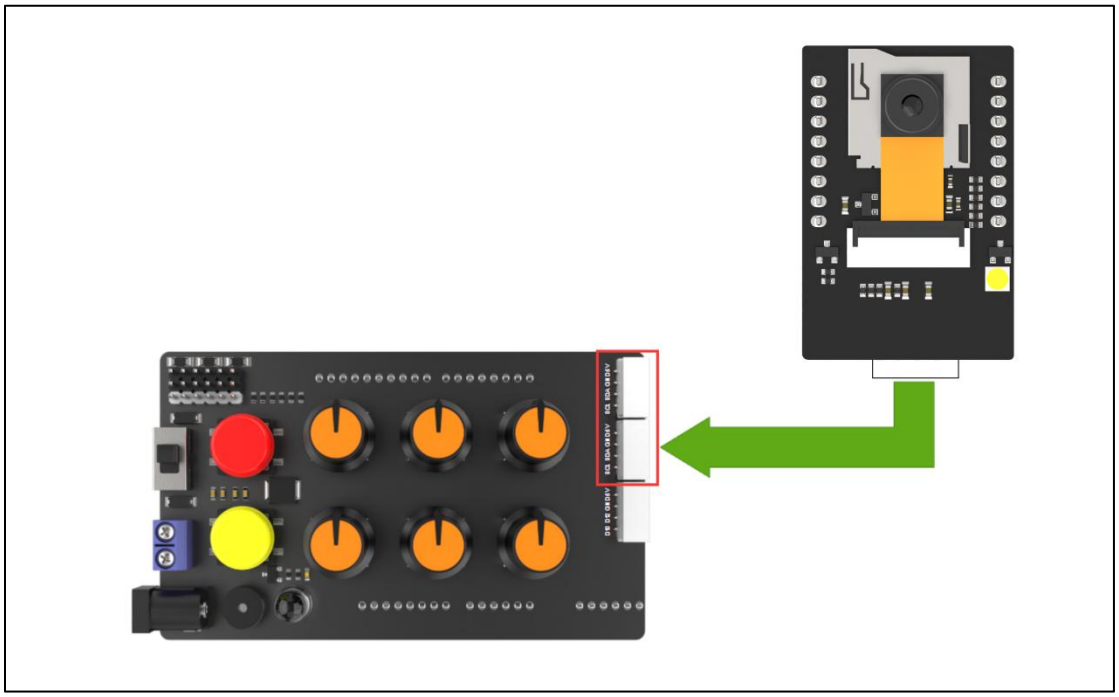


1.3 LAN Mode Connection (Optional)

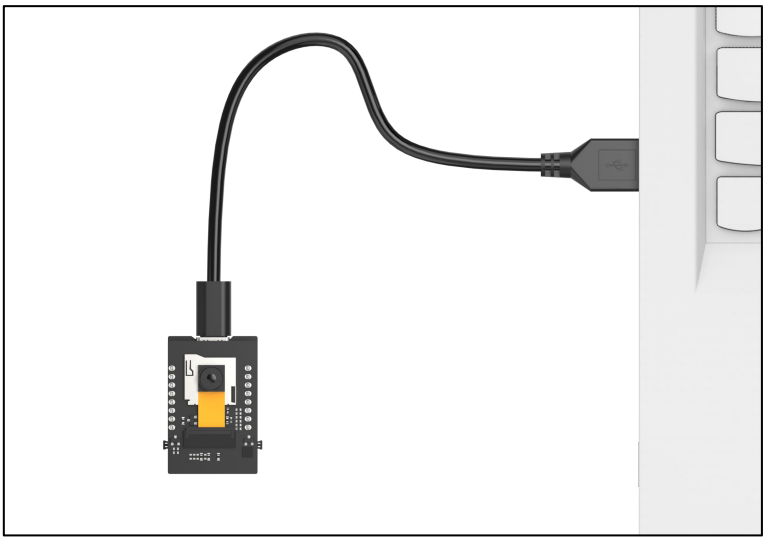
1) Turn on your phone's hotspot and set the name as “HiwonderESP” and the password as “Hiwonder”. Note: the name and password of the hotspot should be same as of this text, otherwise the connection won't succeed.



2) Ensure the camera is connected to the I2C interface on the expansion board, power on the robotic arm and wait for a moment. The camera will automatically connect to the phone's hotspot.

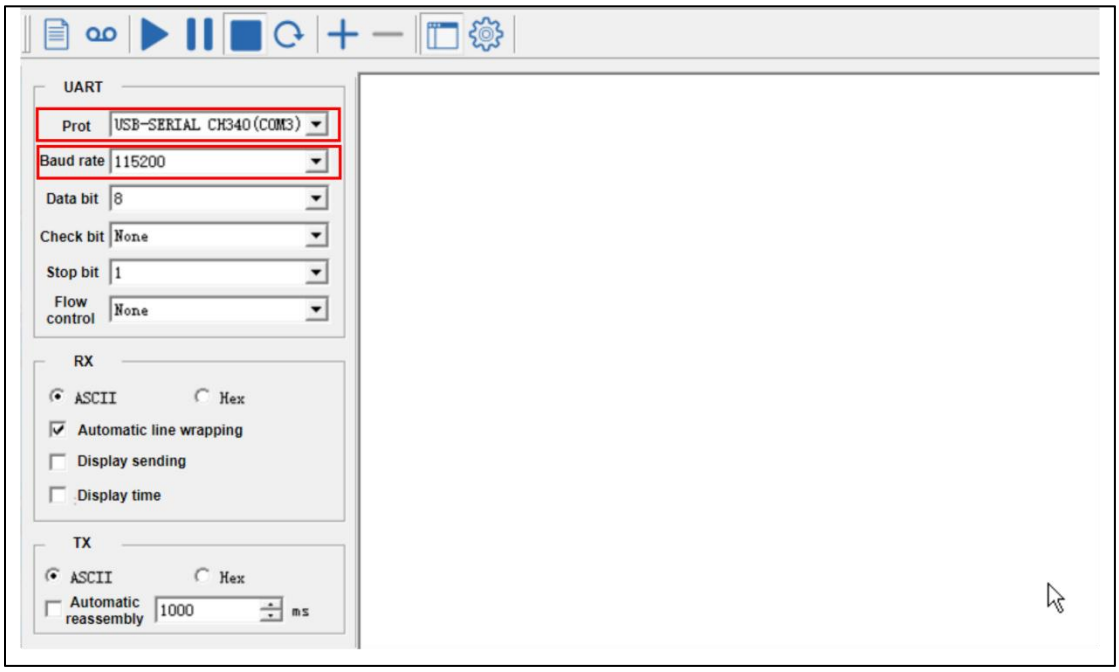


3) After that, connect the camera to the computer with the Type-C cable.

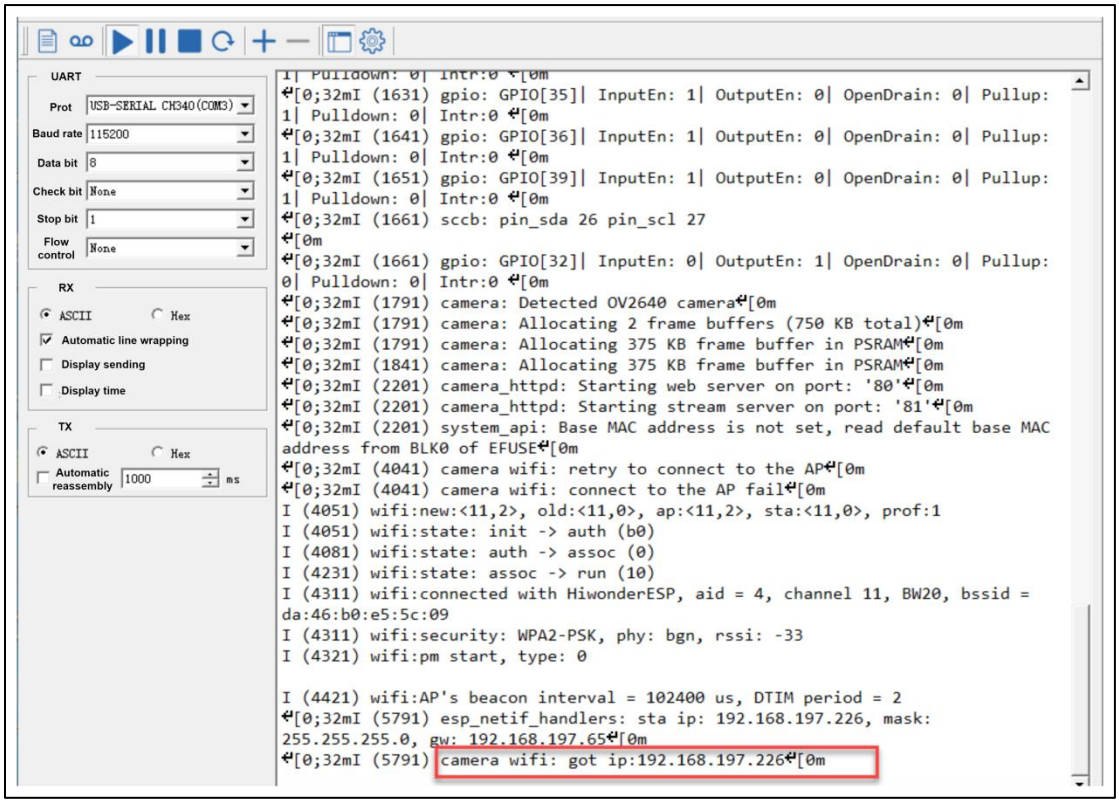


4) Open the serial port utility tool in the path of “2. Software Tools & Program Collections\5. Serial Port Utility Tools”.

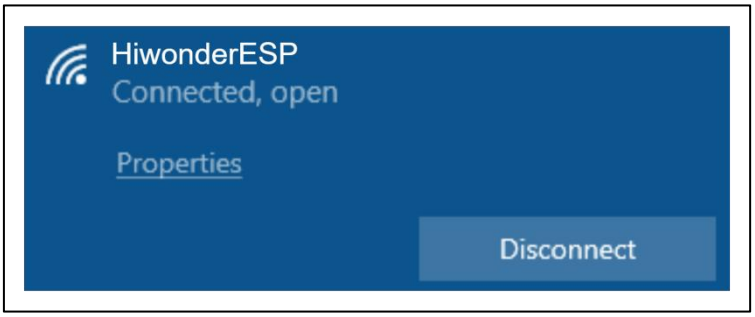
5) Select a port. Take COM3 as example. The port number may vary. Do not choose COM1 which is the system communication port. Set the baudrate as “115200”.



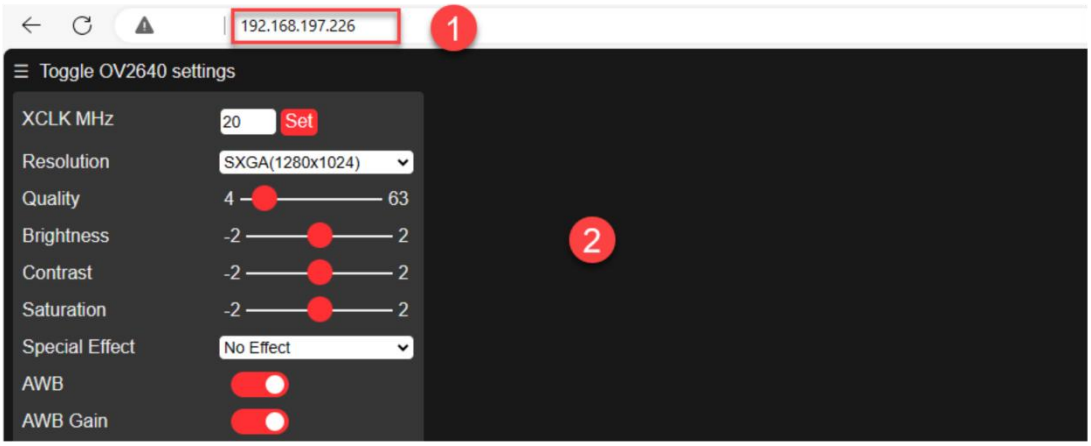
6) Click at the upper left of the page to open the serial port communication. Press the “RST” button on the ESP32-Cam and scroll down to the last line to check your camera's IP address “192.168.197.226”. The IP address may vary, please refer to the actual address.



7) At this time, you can disconnect the Type-C cable and connect the computer to the same hotspot as the camera.

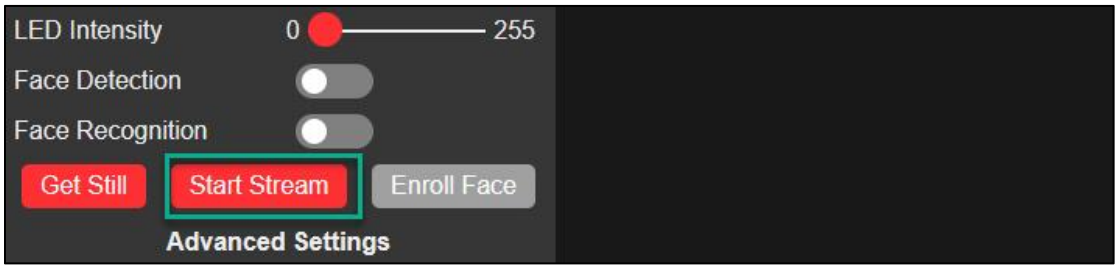


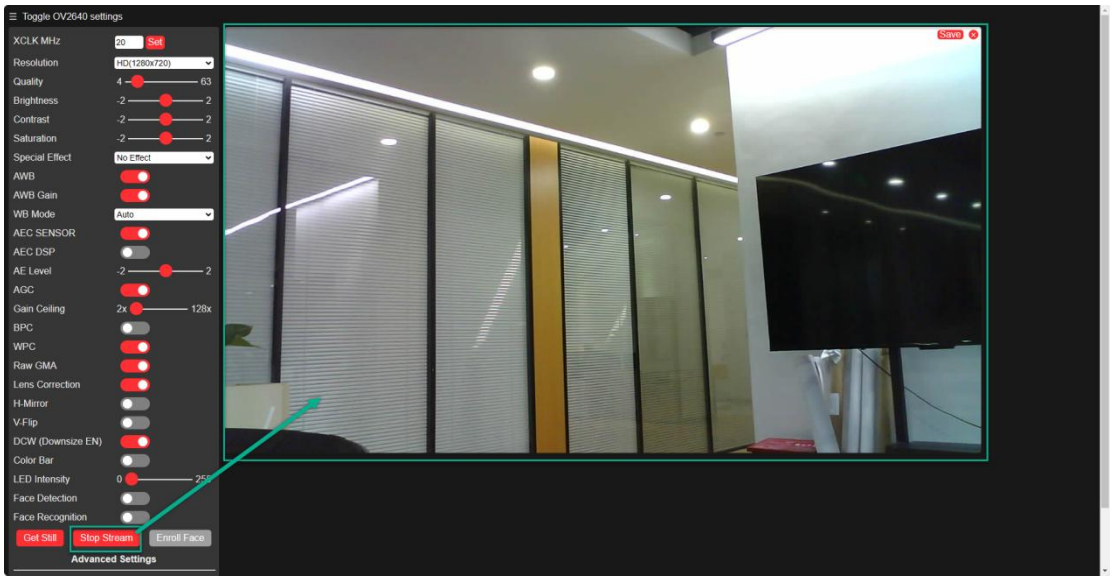
8) In the browser's research bar, enter the IP address of the camera "192.168.197.226" to to access the feedback interface.



2. Start Image Transmission

After the connection is completed, click “Start Stream” in the interface to start image transmission.

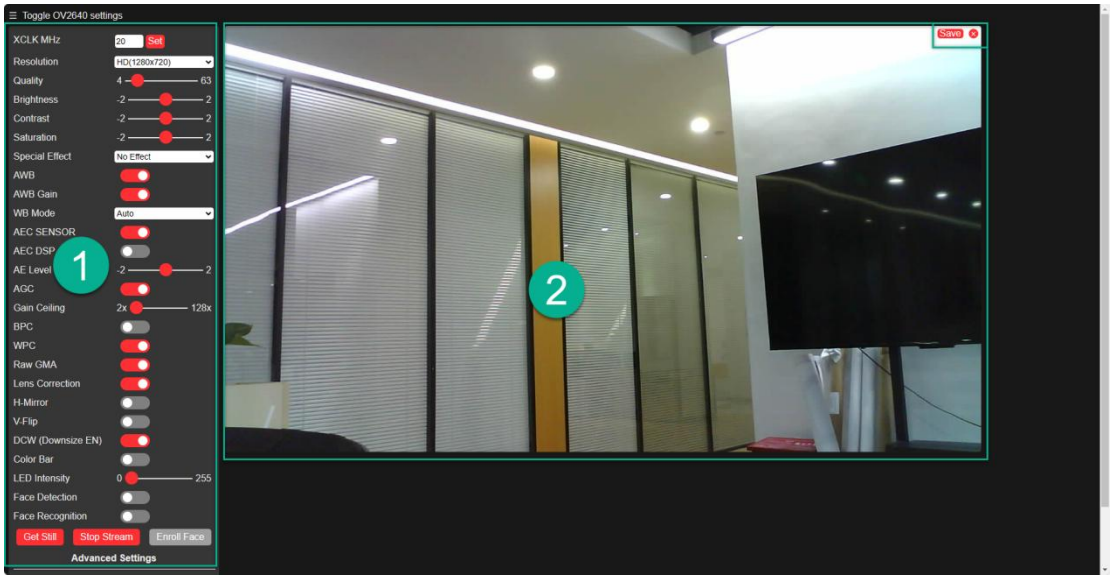




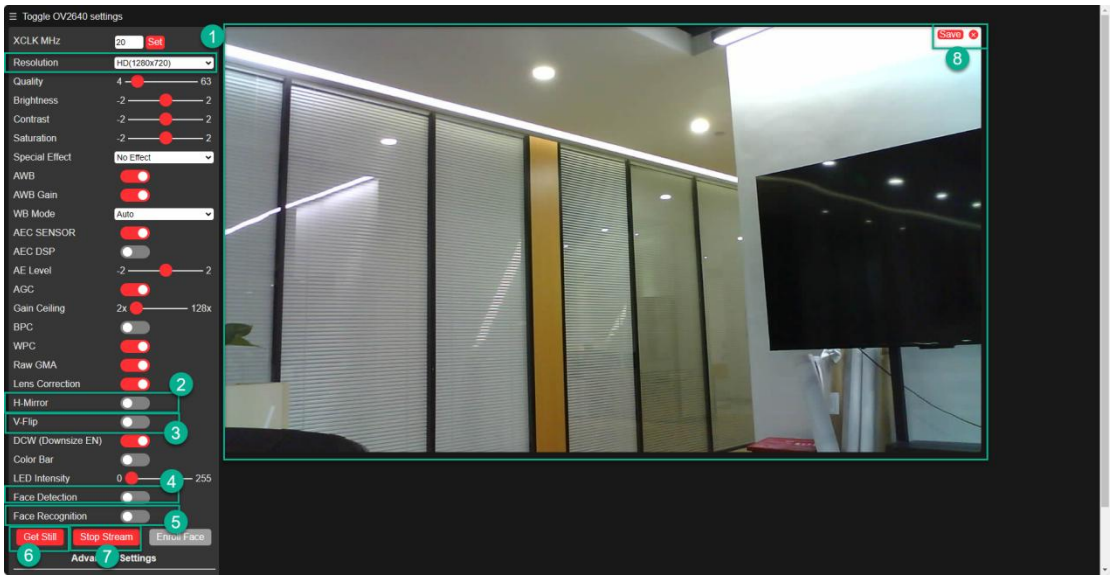
3. Interface Introduction

This interface consists of two parts. One is the setting area (as ①), the

other is the image transmission area (as ②).



The instructions of the common buttons in the interface setting area is as follows:



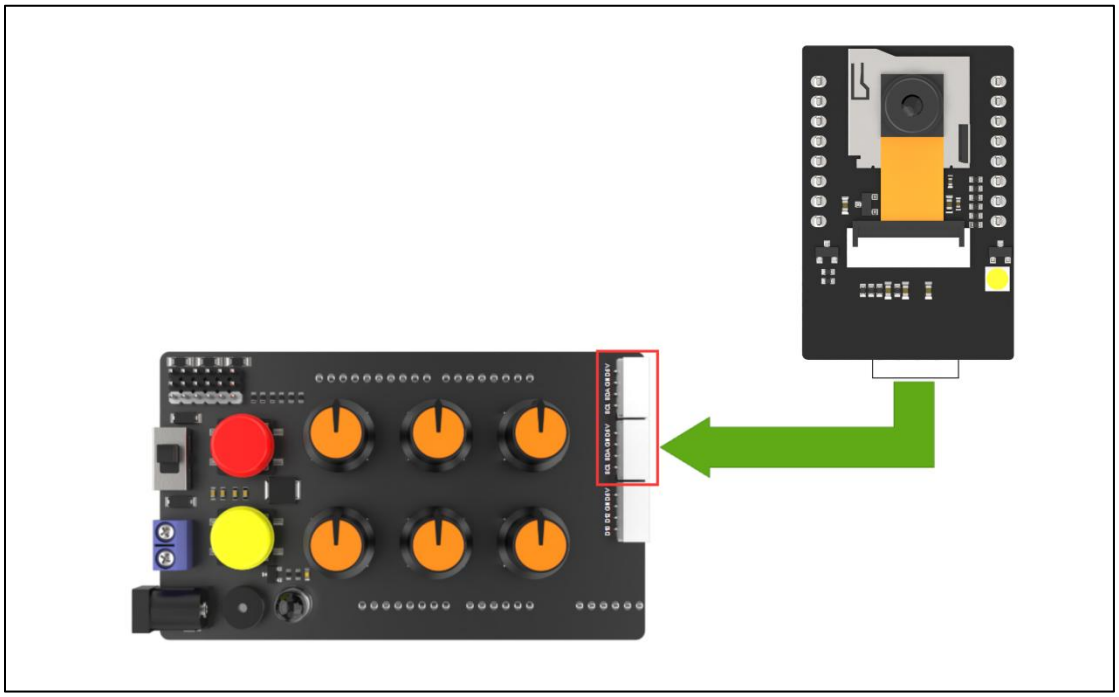
Number	Button	Function
1	Resolution	The resolution of the frame display can be selected. The smaller the resolution is, the smoother the frame will be.

2	H-Mirror	Horizontally flip the frame.
3	V-Flip	Vertically flip the frame.
4	Face Detection	Click it to recognize face.
5	Face Recognition	Choose “Face Detection” and “Face Recognition” buttons at the same time. After the face is recognized, click ‘Get Still’ button to learn specific face.
6	Get Still	Click to learn specific face.
7	Stop Stream	Close Image Postback Frame.
8	Save	Save Images.

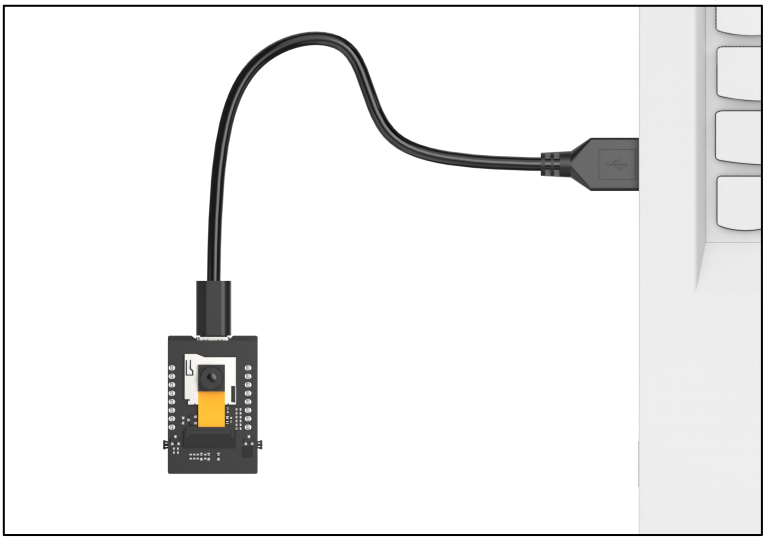
You can access “4.Expanded Tutorials\4.Sensors” to check more detailed ESM32-Cam’s Tutorials.

4. Image Postback Firmware Flashing (Optional)

1) Install and connect the ESP32-Cam to the I2C interface on the MiniArm by accessing “1.Tutorials\6.AI Vision Game Course\6.1 ESP32-Cam Introduction and Installation”.



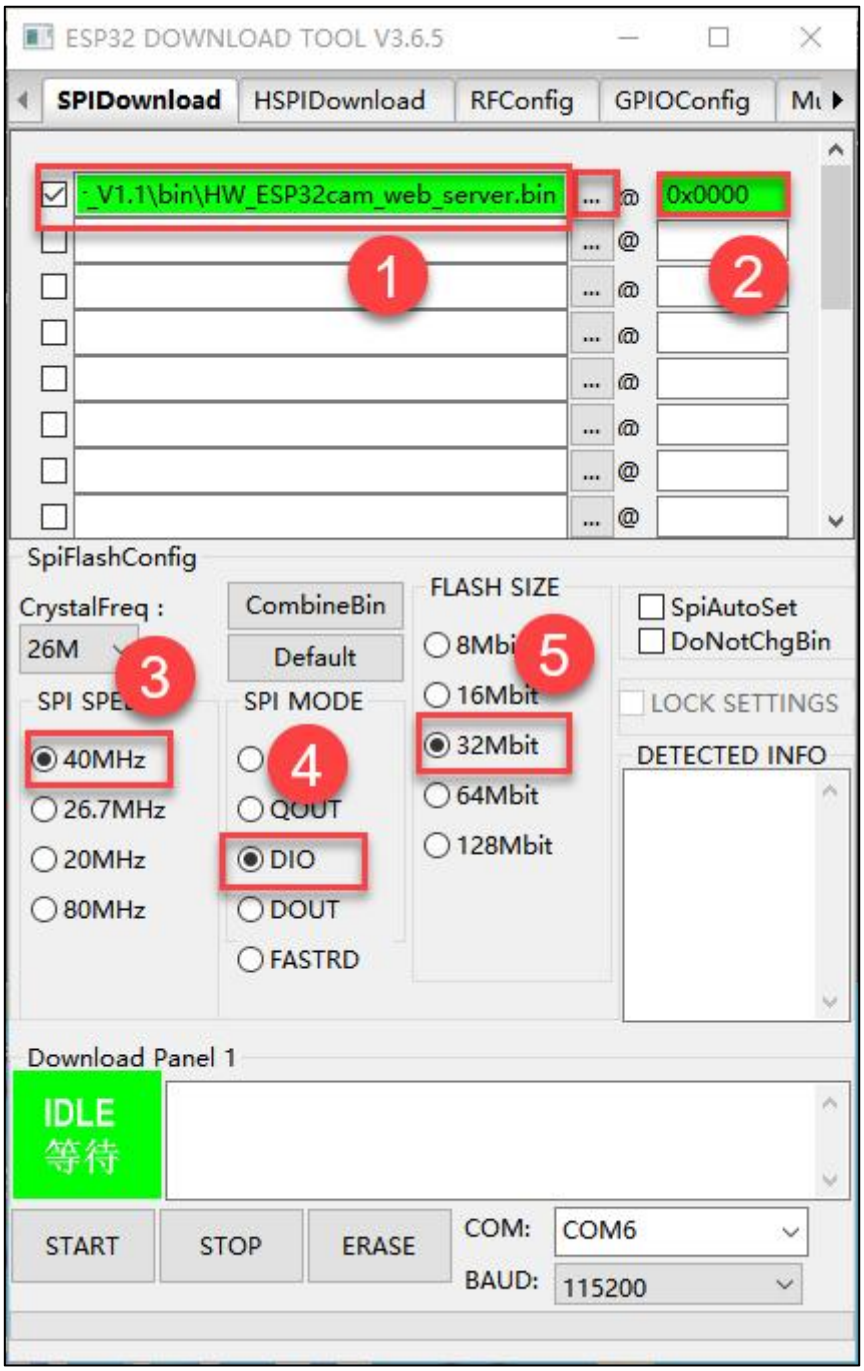
2) Connect the ESP32-Cam to the computer's USB port with a Type-C cable.



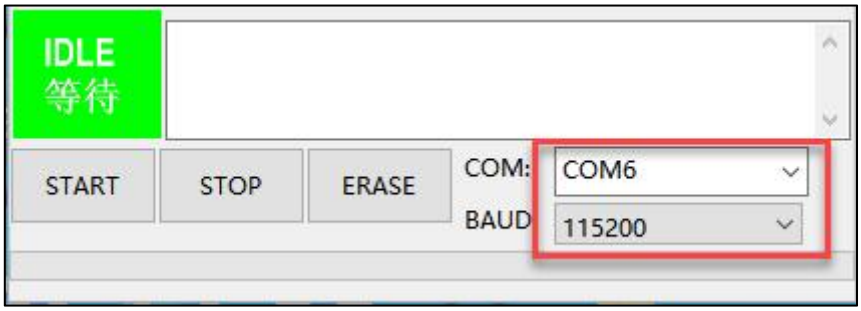
3) Double click and open "flash_download_tools_v3.6.5.exe" file in the "2.Software Tools&Program Collections\4.Firmware Flashing Tools".

4) Then, select "ESP32 DownloadTool" in the opening interface.

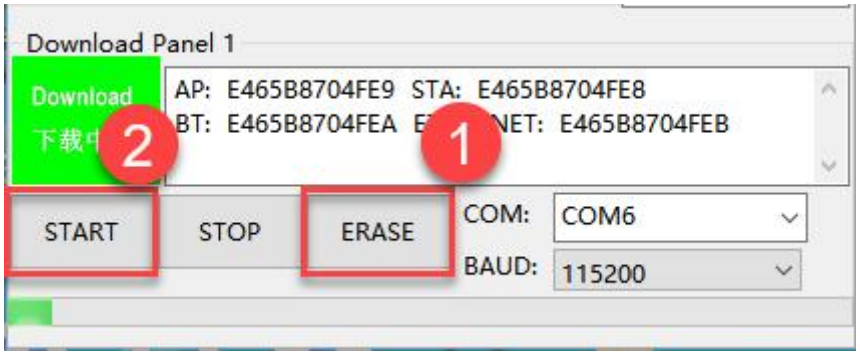
5) In the flashing interface, click to select the firmware first. (Decompress “camera_web_server_V1.1.zip” file to access “camera _web_ server_V1.1\bin\HW_ESP32cam_web_server.bin”.) Next, select the configuration as follows. Note: If the live camera feed appears upside down, please flash the “HW_ESP32cam_web_server_1.bin firmware”.



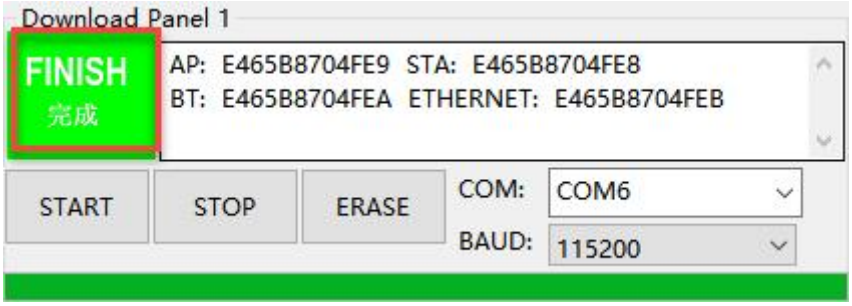
6) Select a port number. The port number relies on the actual situation. Avoid selecting COM1, as it may result in unsuccessful flashing.



7) Click “ERASE” to erase the underlayer firmware. You must do this step first. Next, click “START” to start flash.



8) After a while, the “FINISH” prompt will come out.



9) Then disconnect Type-C cable.